INCH-POUND
MIL-DTL-24643/2E
22 August 2002
SUPERSEDING
MIL-C-24643/2D
22 November 1994

DETAIL SPECIFICATION SHEET

CORD, ELECTRICAL, 300 VOLTS, TYPES LSDCOP AND LSTCOP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated, class K stranding (see table I for size) (see note 2).

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Ethylene propylene rubber or cross-linked polyethylene insulation. Minimum average wall thickness 0.015 inch. Standard identification code applied by method 3.

Fourth - The specified number of conductors (see table I) shall be cabled together with the lay not greater than that specified (see table I). A tie cord shall be laid straight as a central core in LSDCOP-1-1/2. Fillers shall be employed, where necessary, to obtain a firm well rounded assembly.

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Construction (non-watertight) (continued)

Fifth - A polyester binder tape.

Sixth - Cross-linked polyolefin jacket

NOTES:

1. Because of peculiar lay-up of LSDCOP-1-1/2 cord, the centering and circularity requirements shall not apply, provided the minimum thickness requirements are met.
2. At the manufacturer's option, the conductors in type and size LSDCOP-1 may consist of eight strands, each having a nominal diameter of 0.010 inch, around one strand having a nominal diameter of 0.0142 inch.

TABLE I. Details.

Conductors					Overall Diameter			
Military part no. M24643/2	Type and size	No	Size AWG	Cabling lay (max) (inches)	min (inch)	max (inch)	Voltage withstand (min) (volts)	Conductor resistance 1000 feet (max ohms)
-01UN	LSDCOP-1	2	20 (class K)	1	0.235	0.250	1000	11.42
-02UN	LSDCOP-1-1/2	2	18 (class K)	3/8	0.300	0.315	1500	9.00
-03UN	LSDCOP-2	2	18 (class K)	2	0.310	0.330	1500	7.16
-04UN	LSTCOP-2	3	18 (class K)	2	0.325	0.345	1500	7.16

EXAMINATION AND TESTS:

Basic electrical: Requirements:

Conductor resistance - ohms/1000 feet at 25°C, maximum .....(see table I)
Voltage withstand - volts, root mean square, minimum .....(see table I)
Insulation resistance - megohms -1000 feet, minimum 100
Conductor continuity ..... No failure

Group A:

Visual and dimensional..... No failure

Group B:

Cross-linked proof test (percent, maximum)
Insulation 50
Jacket (When tested at 200°C).....50
Tear - pounds per inch thickness, minimum (ASTM D 470) 35
Physicals (unaged)
Insulation
Ethylene propylene rubber
Tensile strength - lb/in^2, minimum 700
Elongation - percent, minimum 250

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Cross-linked polyethylene
Tensile strength - lb/in^2, minimum 1800
Elongation - percent, minimum 250
Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160
Twisting and bending endurance - (see note 3)
At 75°C, cycles, minimum.....6000
At minus 30 ± 2 °C, cycles, minimum.....2000

NOTE: 3. Roller diameters for bending endurance test shall be as follows:

Cable type and size	Roller Diameter (inch, maximum)
LSDCOP-1	0.75
LSDCOP-1-1/2	0.95
LSDCOP-2	1.00
LSTCOP-2	1.00

Group C:
Physicals (aged)
Insulation
Ethylene propylene
Air oven
Tensile strength - percent of unaged, minimum 75
Elongation - percent of unaged minimum 75
Cross-linked polyethylene
Air oven
Tensile strength - percent of unaged, minimum 80
Elongation - percent of unaged minimum 80
Jacket (cord)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum 50
Shrinkage ..... No failure
Permanence of printing (jacket) - cycles, minimum.....125
Heat distortion - percent of unaged, maximum 30

Group D:
Flame propagation (cable) ..... No failure

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable)(125 ± 5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTH: 500 feet (nominal)

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-001)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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